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Does medical students' diagnostic performance improve by observing examples of self-explanation provided by peers or experts?

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Abstract Educational strategies that promote the development of clinical reasoning in students remain scarce. Generating self-explanations (SE) engages students in active learning and has shown to be an effective technique to improve clinical reasoning in clerks. Example-based learning has been shown to support the development of accurate knowledge representations. The purpose of this study was to investigate the effect of combining student's SE and observation of peer's or expert's SE examples on diagnostic performance. Fifty-three third-year medical students were assigned to a peer SE example, an expert SE example or control (no example) group. All participants solved a set of the same four clinical cases (training cases), 1—after SE, 2—after listening to a peer or expert SE example or after a control task, and 3—1 week later. They solved a new set of four different cases (transfer cases) also 1 week later. For training cases, students improved significantly their diagnostic performance overtime but the main effect of group was not significant suggesting that students' SE mainly drives the observed effect. On transfer cases, there was no difference between the three groups ($p > .05$). Educational implications are discussed and further studies on different types of examples and additional strategies to help students actively process examples are proposed.

Keywords Clerkship · Clinical reasoning · Example-based learning · Modeling · Self-explanations

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Introduction

Over the past decades, research has successfully explored multiple facets of clinical reasoning, an essential skill for physicians (Gruppen and Frohna 2002; Norman 2005; Schmidt and Rikers 2007; Eva 2004; Mylopoulos and Regehr 2011). Authors agree that experts' knowledge base is extensive, includes several types of knowledge, and is organised in a way that makes it usable in daily problem solving (Schmidt and Rikers 2007; Charlin et al. 2007; Bordage 2007). The development of clinical reasoning requires time, is critically dependent on exposure to multiple examples and on the repeated practice of solving clinical cases (Norman 2005; Schmidt and Rikers 2007; Kassirer 2010). Clinical exposure alone is insufficient. To develop their knowledge base, medical students must be involved in the problem solving process with opportunities for elaboration and reflection on patient encounters (Schmidt and Rikers 2007; Eva 2004). While clinical rotations provide many opportunities for clinical reasoning and for observing peers', colleagues' and expert clinicians' reasoning (Ajjawi and Higgs 2008; van de Wiel et al. 2011), further strategies are needed to engage students in deliberate clinical reasoning. Self-explanation (SE) appears to be a promising technique in this regard (Chamberland et al. 2011). Another strategy, that of observing examples, has been largely used to support development of accurate knowledge representation in memory (van Gog and Rummel 2010; Rosen et al. 2010; Renkl 2011). In this paper, we explore the usefulness of combining SE and observation of colleagues' SE as learning tools for clinical reasoning.

Self-explanation is a learning technique wherein learners generate for themselves explanations about pieces of information in a text to be learned or a problem to be solved (Chi 1989, 2000, 2011) Through SE, students engage in knowledge elaboration, generation of inferences, integration of new information with prior knowledge, monitoring and consequent knowledge reconstruction or revision of mental representations (Chi 2000; Dunlosky et al. 2013). The positive effects of SE on learning have been shown in a variety of domains (Chi 2000; Dunlosky 2013) including, more recently, in Medicine (Chamberland et al. 2011; Larsen et al. 2013). In an experimental study, students who generated SE while solving clinical cases demonstrated better diagnostic performance 1 week later when tested on different, and more difficult, cases than students who did not self-explain. The positive effect of SE was seen only on cases of a less familiar clinical topic (Chamberland et al. 2011). Studies in other domains also find greater benefit with complex problems and application tasks differing from the initial learning task (Chi 2000; Wong et al. 2002). According to a subsequent related study, lack of familiarity seemed to act as a trigger for medical students to engage in more active thinking while self-explaining for diagnosing cases, as they reactivated more biomedical knowledge to make sense of the patients' findings, possibly creating new links between biomedical and clinical knowledge, thereby constructing a more coherent mental representation of diseases (Chamberland et al. 2013; Woods 2007).

It is relevant for research and educational practice to explore conditions that maximise the effects of SE as a promising technique for developing medical students' clinical reasoning. Combining SE and examples or models of clinical reasoning may provide such a condition. Example-based learning (EBL) is the process of learning by studying examples or observing models performing a skill. The benefits on learning while working with examples is mainly attributed to a lowered cognitive load imposed on students. By providing guidance, examples relieve students of the need to search for every procedural step, allowing them to devote their cognitive capacity to studying and understanding the proposed solution, mainly through SE (van Gog and Rummel 2010; van Merriënboer and

Sweller 2010; Renkl 2014). This in turn helps students construct appropriate knowledge representations of the skill, interrelating abstract principles and concrete examples to guide future performance of that skill (van Gog and Rummel 2010; Renkl 2011).

Example-based learning has been successfully used for a variety of tasks, from highly to less structured, for a wide range of skills, from psychomotor to complex cognitive (van Gog and Rummel 2010). Initial research focused on well-structured domains (e.g., Mathematics) and used mostly printed materials. Recent studies began to investigate more complex examples from less-structured domains (e.g., productive collaboration (Rummel et al. 2009); scientific argumentation (Schworm and Renkl 2007). Clinical reasoning represents a complex and less structured cognitive skill for which EBL could potentially be useful.

For EBL to succeed, the skill to be learned must be observable by the learner. Learning complex cognitive tasks through observation of models requires the latter to externalize cognitive processes and knowledge activated while performing the skill. This can be done by thinking aloud or writing down the actions. SE may represent an alternative way for subjects to make explicit their ongoing clinical reasoning process and relevant knowledge representations. For optimal effectiveness of observing examples, learning design issues must also receive careful attention (van Gog and Rummel 2010; Rosen et al. 2010). The model-observer similarity appears to have a positive influence. A peer model's knowledge, however limited and partial in its scheme, would be in its representation cognitively closer to the observer's own and so would allow the latter to relate to it more easily A coping model in contrast to a mastery model may present errors but also strategies to overcome them which could be helpful for the observer (van Gog and Rummel 2010; Renkl 2014). In contrast, naturally performing experts have extensive knowledge, from which students could benefit, but, on the other hand, having largely automatized their task, work at higher levels of abstraction. Thus, a novice observer may not be able to follow the expert reasoning and learn from it. Still, it remains unclear which similarity features (e.g., competence, background, sex, age) play a critical role in EBL and this issue needs further research. (van Gog and Rummel 2010; Renkl 2014). Another important consideration is the way in which examples are delivered in order to ensure that students engage cognitively and process the provided information while observing (van Gog and Rummel 2010; Renkl 2014). Combining SE with the example appears crucial to optimize its effectiveness. (Renkl 2011; Renkl 2014). Students can also be given the opportunity for problem solving in addition to studying the example. Practice presumably allows students to note deficiencies and help them focus later on relevant aspects of the example.

Both SE and EBL have been shown to contribute to the development of knowledge representations. Clinical reasoning is a complex skill that relies on well-developed mental representations of diseases. The present study examined the added value for learning clinical reasoning of combining clerks' SE with observation of SE examples. Moreover, since the benefit of the model or example appears to depend of its expertise level relative to the student's one, we investigated whether examples from peers or experts would lead to similar or different effects on diagnostic performance. We hypothesized that observing explicit examples would give additional opportunities for students to extend their understanding and critically revise their own knowledge representations, and that this would translate into better diagnostic performance. We also hypothesized that because of their cognitive proximity; peer models would be at least as effective for learning as expert models.

Methods

Design

This mixed-design experiment comprised two phases. In the training phase, participants were assigned either to the peer-SE group, to the expert-SE group, or to the control group. They solved a set of 4 clinical cases twice: after having self-explained and then again after having listened to an example of SE from a peer or an expert (peer- or expert-SE groups), or without listening to any example (control group). In the assessment phase, 1 week later, all participants had to solve the same 4 cases again (training cases) and a new set of four more difficult cases (transfer cases). Figure 1 summarizes the study design.

Participants

Participants were 53 third-year medical students from the *Université de Sherbrooke* in Québec, Canada. The undergraduate program comprises 4 years, with 2 ½ years of PBL followed by 18 months of clerkship. Students in their first 8 months of clerkship, in two consecutive cohorts (academic year 2011–12, and academic year 2012–13), were invited by email to participate in the study. Participation was voluntary, without financial incentive. Written informed consent was obtained. Ethics approval was obtained from the *Comité d'Éthique de la Recherche, Éducation et Sciences Sociales* of Université de Sherbrooke.

To reach the required number of participants, we recruited them in two consecutive cohorts. In the first cohort, 151 students were approached for the study. Forty students accepted and were randomly assigned to the peer or to the expert SE group for the training

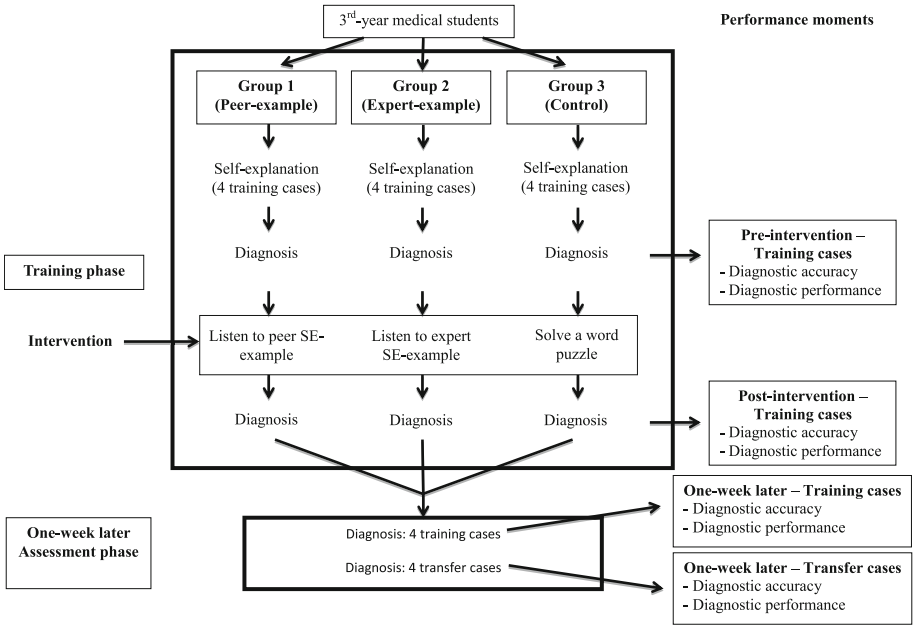

Fig. 1 Study design

phase. Thirty-seven completed the assessment phase (peer-SE group = 18; expert-SE group = 19). In the second cohort, 21 students were invited to participate. Eighteen students accepted the invitation and 16 completed the two phases of the study. These students were assigned to the control group ($n = 16$).

Materials

Eight written clinical cases, all addressing the topic of jaundice were used for this study. Jaundice was a rather unfamiliar topic to the 3rd-year medical students involved (Chamberland et al. 2011). The topic was broken down into four broad categories of diseases with different pathophysiological mechanisms: acute hepatitis, chronic liver disease, obstructive jaundice and hemolysis. A set of four cases, one per category, was used in the training phase (training cases). In the assessment phase, the training cases were again used as well as a second set of four different cases (transfer cases), one per category again, but with different diagnoses from the training cases (see "Appendix 1").

Case descriptions of approximately 300 words including chief complaint, background information, and findings from history, physical examination and laboratory tests were developed and used in a previous study (Chamberland et al. 2011).

In the training phase, two versions of booklets, each with a different case order, were distributed alternately to the recruited students. Similarly, in the assessment phase, clinical cases were presented in a booklet using again two differently ordered versions. The training cases were presented first, followed by the transfer cases with three intercalated filler cases on unrelated clinical topics.

As peer and expert SE examples, we used audio-recorded SE. To reflect the variety of examples possibly encountered in the clinical environment, SE examples from three different peers and three different experts in Internal Medicine, were prepared as follows. In individual sessions, clinical clerks and internists, not involved in the study, were first introduced to SE, and then asked to self-explain on each of the four training cases. They were asked not to provide verbally their final diagnosis at the end to avoid that students who would listen to the examples would hear the final correct response. Experts were specifically instructed to self-explain at their level of expertise rather then think or talk as clinical teachers. All audiotapes were anonymous. Their speed was adjusted to create equally long (8 min) audio examples. Written consent was obtained from peers and experts.

Expectably, there were notable differences between the peers' and the experts' SE examples. Experts generated very early a problem representation including key elements for one or a few likely diagnoses. Additional clinical data in the case were used mainly to confirm early diagnoses or quickly rule out alternatives. Experts' examples contained almost no explicit reference to underlying mechanisms, no errors, almost no ambiguities or hesitations, and very few paraphrases. In contrast, peers mainly analysed isolated clinical data, did not quickly recognize key elements and considered all along numerous and varied diagnoses. They referred to underlying mechanisms, made some erroneous interpretations, raised questions, expressed hesitations and used frequent paraphrases.

Procedure

Training phase

A research assistant tested all participants individually. Students were first provided with a general definition of SE and then listened to a 7-min audio example of SE on a clinical case

(mono-arthritis) not used in the study but with a similar format. Subsequently, they were asked to read each case, generate SE out loud for 8 min, and take 2 min to write down their responses to the following three questions:

Q1. What is the most likely diagnosis?
Q2. What are the two main arguments supporting this diagnosis?
Q3. List two plausible alternative diagnoses.

The first question provided a measure of diagnostic accuracy; combination of the three questions was intended to provide a rough measure of the quality of the diagnostic reasoning. We refer to this first performance moment hereafter as "pre-intervention".

Participants of the peer-SE and expert-SE groups were then instructed to listen to an 8-min recording of either peer or expert SE, and thereafter to answer again the same three questions in 2 min. As to the control group, in an effort to prevent further SE on their part, students were invited to move to the next page of their booklet to solve a word puzzle for 8 min and then answer again the three questions. We refer to this second performance hereafter moment as "post-intervention". This procedure was repeated for all four training cases. Each participant was exposed to a random sample from all of the three peers or the three experts to reflect the variety of examples in the real learning environment. Time was strictly controlled. The training phase lasted 1.5 h.

Assessment phase

One week later, participants were met individually for the assessment phase, which was identical for the three groups. Students had not been informed in the training phase about the tasks they would have to carry out during the assessment phase. Participants were first asked to read the four cases used in the training phase (training cases), generate SEs, and answer again the three questions regarding their diagnosis. Second, participants were asked to solve in silence the four transfer cases with three filler cases interpolated. Time to solve each case was not controlled but participants were allowed up to 2 h for the whole assessment phase. We refer to this third performance moment hereafter as "1-week later".

At the end of the assessment phase, students completed a brief questionnaire about potentially relevant learning opportunities (i.e., current clinical rotation, individual study time, and teaching activities related to jaundice) during the week of the study. The different groups reported similar opportunities. Four research assistants ran all the sessions, after having gone through a standardization exercise, to ensure a uniform procedure.

Data analysis

A scoring grid, developed and validated in a previous study (Chamberland et al. 2011), was used to evaluate students' answers to each question. Each answer was scored as 0, if it was incorrect, 1, if partially correct, and 2, if entirely correct. Two scores were derived from participants' answers: the diagnostic accuracy score (the score on question Q1) and the diagnostic performance score (the sum of scores on questions Q1, Q2 and Q3). For each participant, the scores of the four cases (either training or transfer cases) were summed. The scores were then expressed as a percentage of the maximum score (8, for the diagnostic accuracy, and 24 for the diagnostic performance). One rater (MC) scored the responses of all students from all groups. In the previous study, two independent raters scored students' responses. Inter-rater reliability, measured by intra-class correlations, had

ranged from 0.70 to 0.95 (Chamberland et al. 2011). Compared to students' responses from the previous study, no new responses were provided by participants in the present study.

Descriptive statistics were computed. To assess the comparability of the groups, two separate one-way ANOVAs were performed on the mean scores obtained for diagnostic accuracy and diagnostic performance before the intervention. For the training cases, two separate repeated-measure ANOVAs with group (peer SE, expert SE and control) as a between-subjects factor, and performance moment (three performance moments: before and after the intervention in the training phase and 1 week later in the assessment phase) as a within-subjects factor were performed on mean diagnostic accuracy scores and diagnostic performance scores. For transfer cases in the assessment phase, two separate one-way ANOVAs with group as a between-subjects factor were performed on mean diagnostic accuracy scores and diagnostic performance scores. Significance was set at 0.05.

Results

Tables 1 and 2 present mean scores obtained, respectively, on diagnostic accuracy and diagnostic performance as a function of performance moment and experimental condition on training cases. No significant differences between the groups were found before the intervention, either in diagnostic accuracy score ($F[2, 50] = 1.06$, $p = .354$) or in diagnostic performance score: ($F[2, 50] = 0.19$, $p = .828$).

For the diagnostic accuracy score, Mauchly's test indicated that the assumption of sphericity had been violated, X^2 (2) = 12.66, $p = .002$, therefore, the multivariate tests are reported ($\in = .82$). The repeated-measure ANOVA revealed a significant main effect of performance moment ($V = 0.37$, $F[2, 49] = 14.45$, $p < .001$). The main effect of group was not significant ($F[2, 50] = 0.66$, $p = .520$), but there was a significant interaction effect ($V = 0.23$, $F[4, 100] = 3.30$, $p = .014$). To better understand these primary findings, post hoc analyses were conducted using paired t-tests within each group (with Bonferroni correction). The p-values are presented in Table 1. In both the peer- and the expert-SE groups, students' diagnostic accuracy scores improved significantly immediately after listening to the SE example ($p = .007$, $p < .001$ respectively), and then remained unchanged 1 week later ($p = .298$, $p = 1$ respectively). The control group showed a borderline but delayed improvement, i.e., 1 week later ($p = .022$). Three One-way ANOVAs on mean diagnostic accuracy scores at each performance moment were also conducted as post hoc analyses. A significant difference between the groups was noted only immediately after the intervention ($F[2, 50] = 3.29$, $p = .045$. This difference was observed between the expert SE group and the control group ($p = .041$).

For the diagnostic performance scores, Mauchly's test also indicated that the assumption of sphericity had been violated (X^2 [2] = 12.44, $p = .002$) therefore, the multivariate tests are reported ($\in = .82$). A significant main effect of performance moment was observed ($V = 0.46$, $F[2, 49] = 21.27$, $p < .001$). The main effect of group was not significant ($F[2, 50] = 0.30$, $p = .744$) and there was no significant interaction effect ($V = 0.12$, $F[4, 100] = 1.60$, $p = .179$). *Post hoc* analyses using paired *t*-tests within each group are presented in Table 2. The diagnostic performance scores in the peer and the expert-SE groups improved significantly immediately ($p = .002$, $p < .001$ respectively) and remained stable 1 week later ($p = .622$, $p = .607$ respectively). For the control group, the immediate improvement was borderline significant ($p = .021$) and remained stable ($p = .525$).

Table 1 Mean diagnostic accuracy scores (*SD*) as a function of performance moment for each group on training cases and *p* values for score differences between performance moments for each group

Group (*n*)	Pre[a]	Post[b]	One-week[c]	Pre versus post	Pre versus one-week	Post versus one-week
Peer SE (18)	70.83 (16.61)	82.64 (9.72)	85.42 (9.82)	.007	.004	.298
Expert SE (19)	74.34 (13.48)	87.50 (9.32)	87.50 (9.32)	.000	.003	1.000
Control (16)	78.13 (13.31)	78.13 (13.31)	86.72 (13.28)	1.000	.022	.022

[a] Pre-intervention

[b] Post-intervention

[c] 1-week-later intervention

Significance level $p = .017$

Table 2 Mean diagnostic Performance Scores (*SD*) as a function of performance moment for each group on training cases and *p* values for score differences between performance moments for each group

Group (*n*)	Pre[a]	Post[b]	One-week[c]	Pre versus post	Pre versus one-week	Post versus one-week
Peer SE (18)	60.88 (13.19)	70.37 (9.99)	71.53 (8.84)	.002	.008	.622
Expert SE (19)	60.09 (9.03)	70.61 (6.73)	69.74 (8.08)	.000	.001	.607
Control (16)	62.50 (12.64)	66.15 (13.25)	67.45 (8.90)	.021	.047	.525

[a] Pre-intervention

[b] Post-intervention

[c] 1-week-later intervention

Significance level $p = .017$

Table 3 Mean diagnostic accuracy scores (*SD*) and diagnostic performance scores (*SD*) for each group on Transfer cases 1 week later

Group	Mean diagnostic accuracy scores	Mean diagnostic performance scores
Peer SE	56.25 (17.81)	58.56 (11.39)
Expert SE	60.53 (21.76)	59.87 (13.41)
Control	53.91 (20.78)	58.85 (10.74)

Table 3 presents mean diagnostic accuracy scores and mean diagnostic performance scores on transfer cases for each group. No significant differences emerged between the groups either in diagnostic accuracy scores ($F[2, 50] = 0.49$, $p = .616$) or in diagnostic performance scores ($F[2, 50] = 0.06$, $p = .941$).

Discussion

The present study investigated the impact on students' diagnostic performance of listening to peer or expert SE examples after having themselves self-explained while diagnosing clinical cases. We assessed the short term and long-term (1-week) effect on the training cases and the long-term effect on transfer cases. It was hypothesized that listening to SE examples would add to students' SE and support their knowledge construction thus translating into better subsequent performance.

The results showed that, for the training cases, students' diagnostic accuracy and diagnostic performance improved significantly across the study phases relative to their initial performance, whether or not it was followed by listening to SE models. A closer look at the changes over time within each group leads to interesting additional observations. First, differences in the magnitude of improvement suggests that there was in fact some added value of being exposed to SE models. Combining SE models to students' own SE led to significant improvement, while the improvement with students' SE alone was only borderline. Second, the time frame of the improvement differed. Combining SE models with students' own SE led to an immediate improvement in diagnostic accuracy that was maintained after 1 week, while improvement of students' SE alone was delayed, only showing up in the assessment phase. SE examples presumably provide some form of immediate feedback to students, which may explain why they improved already in the training phase. On the other hand, the delayed positive effect of students' own SE noticed in the control group suggests that SE alone is useful for learning. However, since the underlying mechanisms presumably involve knowledge elaboration and restructuration, this may take time and may not be visible in immediate testing but showed up later, on the 1-week measurement. This is in line with other studies on medical students' clinical reasoning in which the effect of deep processing of the information either by elaboration or structured reflection requires time and may even lead to poorer performance initially, followed by significant improvement later on (Mamede et al. 2012; Woods et al. 2007a, 2007b). Third, the peer SE and the expert SE groups' outcomes were identical. While SE models led to some additional improvement, the type of SE model (peer vs. expert) did not make any difference on students' performance, that is, peers' examples were as good as experts' examples.

For transfer cases 1 week later, no added value of SE models was noted. For this task that reflects deeper learning than memorization and retention (Mayer, 2010), students' SE alone was as effective as the combination with SE examples.

Thus the added value of SE models was altogether subtle, only evidenced on training cases, and expert SE models were not more effective than peer SE models. Apparently, students' own SE was in large part responsible for the overall improvement across all groups, adding further evidence to the positive influence of this learning technique in Medicine (Chamberland et al. 2011; Larsen et al. 2013; Chamberland et al. 2013).

Why do SE models not add much to students' performance? The explanation may lie in the nature of the examples used in this study. In the worked-example literature, examples are usually ideal ones, didactically designed and presenting only steps relevant to the task (van Gog and Rummel 2010). In contrast, in our study, we used natural "raw" examples of SE, as they might occur in clinical setting. We deliberately chose to use non-didactically designed examples of clinical reasoning to be able to transpose more easily the results of the study to the clinical settings relevant to clerkship. The level of expertise of the example may also play a role. We used examples from peers (third-year medical students), whose SE included irrelevant details and even uncorrected errors. Despite their cognitive proximity, participants at the exact same level may have little to learn from such examples. As to the expert SE examples, while it might be counter-intuitive that students would not learn from an expert's "flawless" clinical reasoning, experts behaving naturally may not be the best examples for novices (van Gog and Rummel 2010; Rosen et al. 2010). Experts use fast, largely non-analytic reasoning, tend to skip steps and to reason at a higher level of abstraction (Bordage 2007), which may be difficult for students to relate to their own knowledge.

Another possible explanation for the limited impact of SE models might be related to the degree of students' cognitive involvement while listening to the examples. Participants were informed that they would listen to an example of SE from someone else and that they would have to solve the problem again. They were not given prompts or other specific tasks to ensure that they listened actively to the examples. They may have been passive, even though their prior SE on the same case was expected to act as a preparatory task that would help them to focus on the example that followed. Nevertheless, perhaps more is needed to keep students actively engaged with the example. In a recent article proposing an instructionally oriented theory of example-based learning, Renkl (2014) emphasized the critical role of engaging students using SE *while* working with the example.

Finally, clinical reasoning is a complex cognitive task, in contrast to more structured tasks and procedural skills for which models and examples have been more often used (van Gog and Rummel 2010; Rosen et al. 2010). Measuring an impact of observing examples may be a matter of dosing, meaning that examples may have to be multiple, varied, repeated and spaced over time.

This study has limitations. Students in the control group were recruited consecutively and were not randomized in contrast to students in the peer- and expert-SE groups. However, all participants were at the same level of training from the same program. We used SE examples of peers (third year medical students) and experts, and the results may differ if examples of intermediate levels of expertise (e.g., residents) are tested.

Finding ways of keeping students cognitively active during the clerkship, and designing activities that maximise learning from unplanned activities in clinical settings remain a challenge for clinical teachers and educators. While recent studies suggest an important role of observing colleagues for the development of clinical reasoning in the workplace (Ajjawi and Higgs 2008: van de Wiel et al. 2011), little is known on how to maximise

learning from these naturally occurring models. In fact, our findings advise cautious in subscribing to the widespread view that having students listening to expert clinicians reasoning aloud while solving clinical cases is beneficial for their learning. SE seems to be a promising individual learning tool for clinical reasoning in clerkship. However, there seems to be limited value of adding SE examples, at least examples from peers or experts that are natural, without further instructions to students to actively process these examples. Even though this study showed partly negative findings, it has potentially important implications since medical students in the clerkship spend significant amounts of time listening to examples of clinical reasoning from peers or experts that are even more implicit than the ones used in the present study. We should find better ways to seize learning opportunities from observing others. Future studies should examine the impact of SE examples of other levels of expertise and investigate additional strategies that could help students actively process and use natural examples of clinical reasoning.

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Appendix

See Table 4.

Table 4 Template for clinical cases development

Broad category of diseases	Specific diagnosis	
	Training cases	Transfer cases
Acute hepatitis	Viral hepatitis B	Viral hepatitis A
Chronic liver disease	Chronic alcoholism	Hemochromatosis
Obstructive jaundice	Pancreatic tumor	Choledocholithiasis
Hemolysis	Auto-immune, idiopathic	Cold agglutinins, Mycoplasma infection

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